



Tool Calling Experiment Analysis Report

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Experiments analyzed: 105 experiments across 5 methodologies



Executive Summary

Total Experiments

105

Total Test Cases

1,050.0

Best Accuracy

100.0%

fill_adaptive_10tools

Fastest Avg. Latency

389ms

phase1_mcp_10tools_medium



Token Usage Summary

Experiments with Token Data

105

Avg Input Tokens

4,750

Avg Output Tokens

53

Most Token-Efficient

607

fill_hybrid_10tools



Methodology Rankings (by Average Accuracy)

#1 Adaptive RAG: 93.1%

#2 RAG: 92.0%

#3 Hybrid: 86.9%

#4 MCP (Baseline): 67.7%

#5 Clustering: 45.5%

Note: Rankings are calculated treating missing experiments as 0% accuracy. This penalizes methodologies that could not be tested at certain tool scales (e.g., due to token limits).

Missing experiments:

- MCP (Baseline): 500, 985 tools

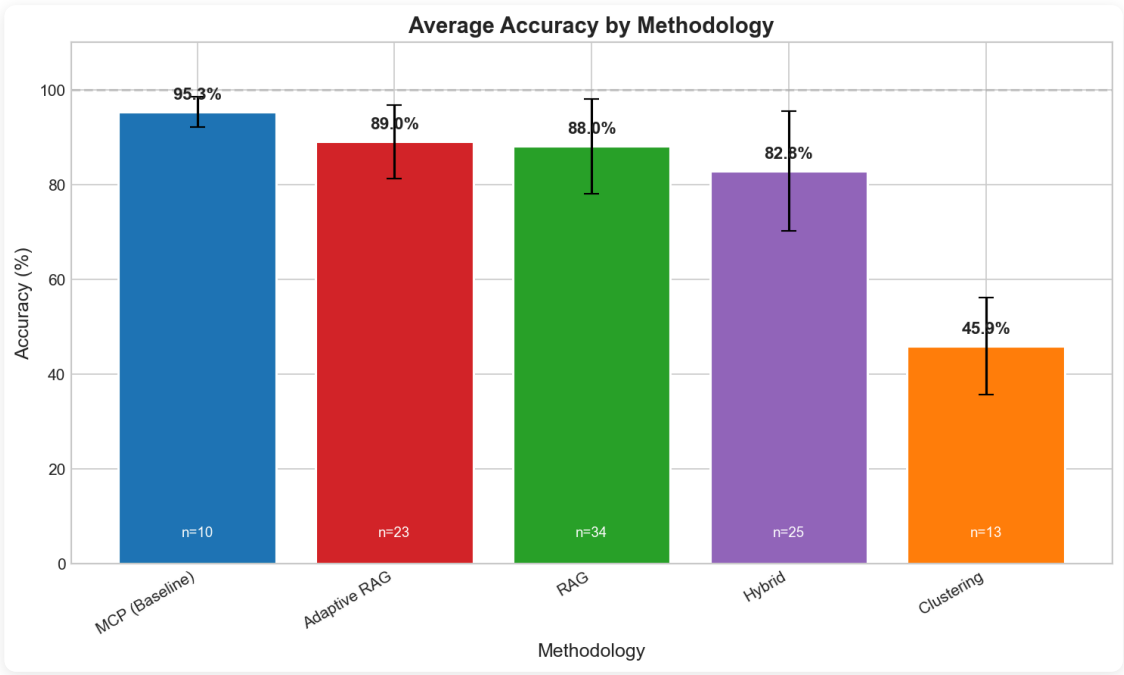


Summary Statistics

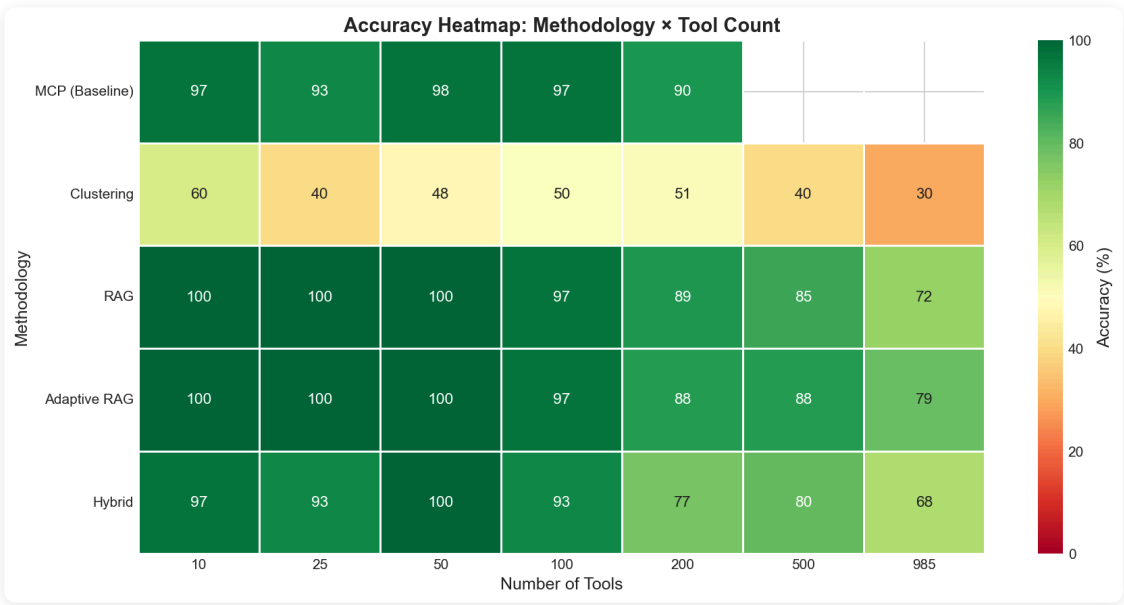
	Accuracy (Mean)	Accuracy (Std)	Accuracy (Min)	Accuracy (Max)	Latency Mean (ms)	Latency Min (ms)	Latency Max (ms)	Total Tests	Num Experiments
methodology									
adaptive_rag	0.890	0.077	0.733	1.000	3552.804	739.472	7089.247	230.0	23
clustering	0.459	0.102	0.300	0.667	2330.524	780.550	12879.243	130.0	13
hybrid	0.828	0.127	0.533	1.000	2812.438	424.892	7334.664	250.0	25

	Accuracy (Mean)	Accuracy (Std)	Accuracy (Min)	Accuracy (Max)	Latency Mean (ms)	Latency Min (ms)	Latency Max (ms)	Total Tests	Num Experiments
methodology									
mcp	0.953	0.032	0.900	1.000	7378.808	388.891	25231.272	100.0	10
rag	0.880	0.100	0.633	1.000	3401.604	779.037	7866.622	340.0	34

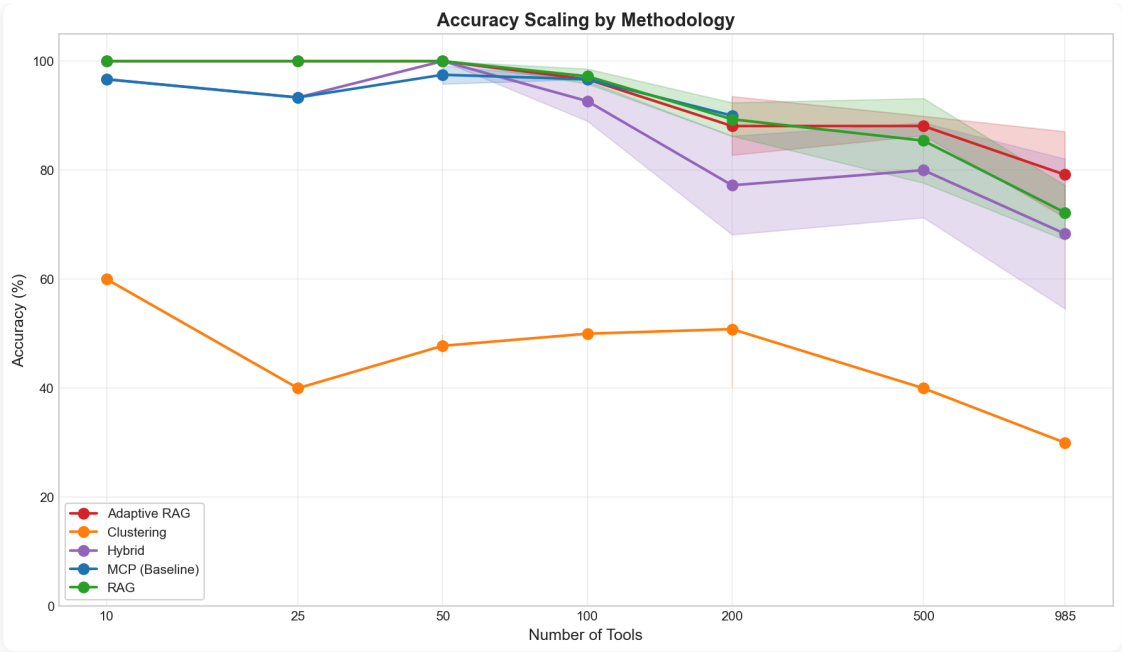
Visualizations



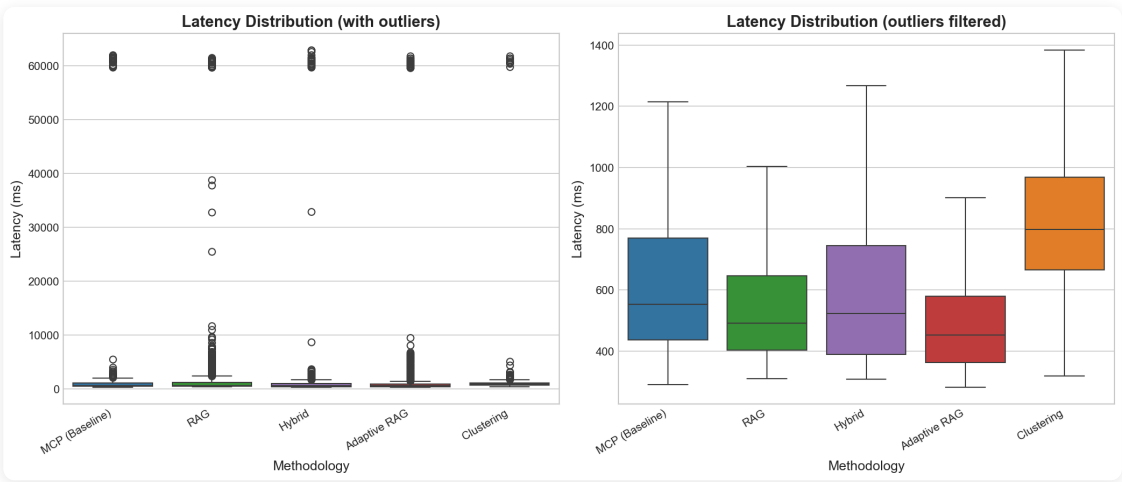
Average accuracy comparison across methodologies with standard deviation error bars.



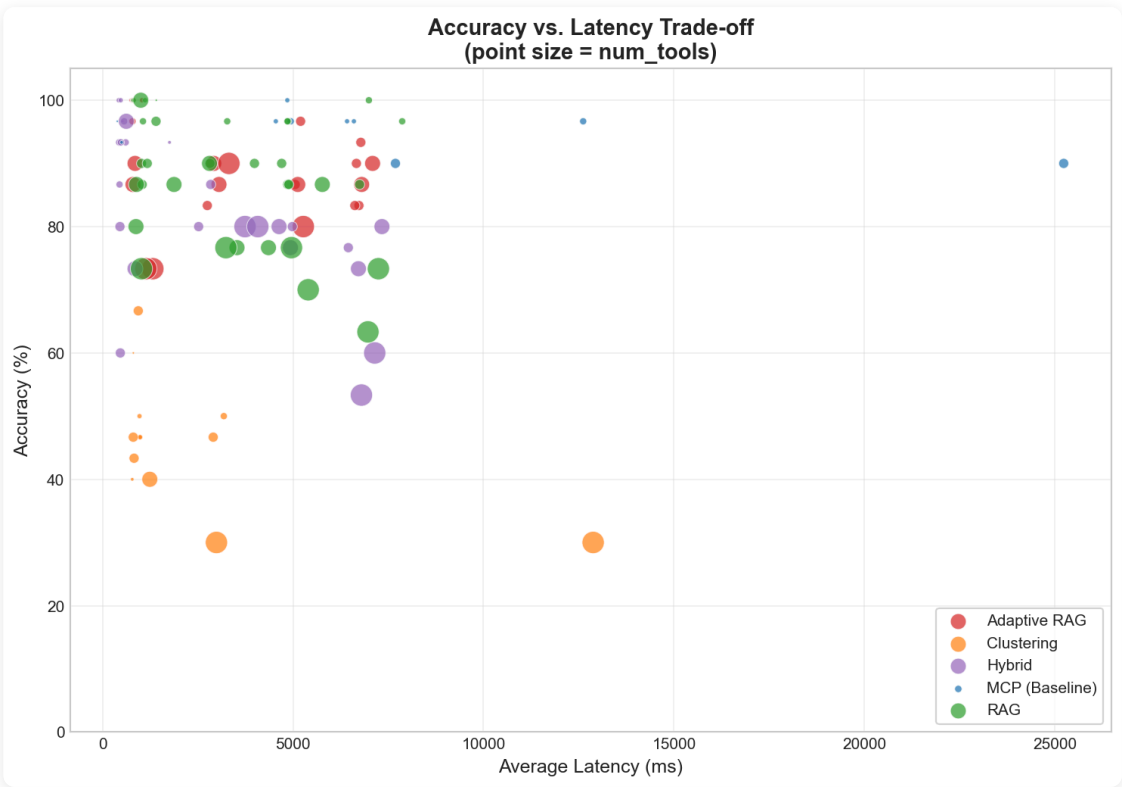
Accuracy by methodology and tool count. Green = high accuracy, red = low accuracy.



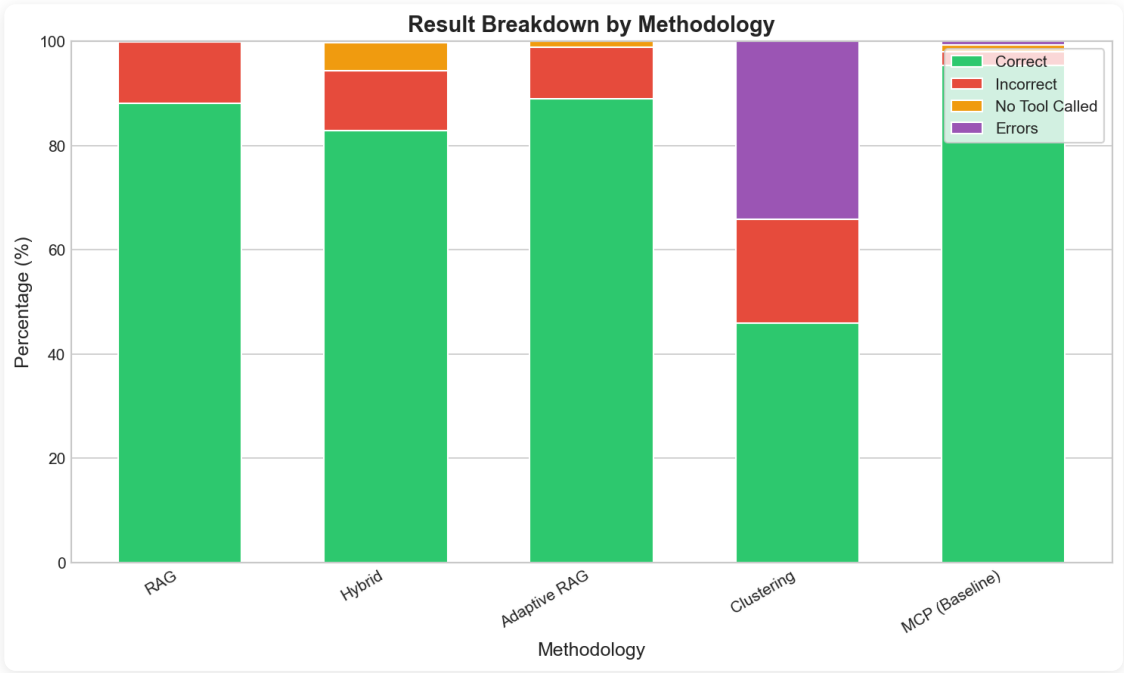
How accuracy changes as the number of tools increases for each methodology.



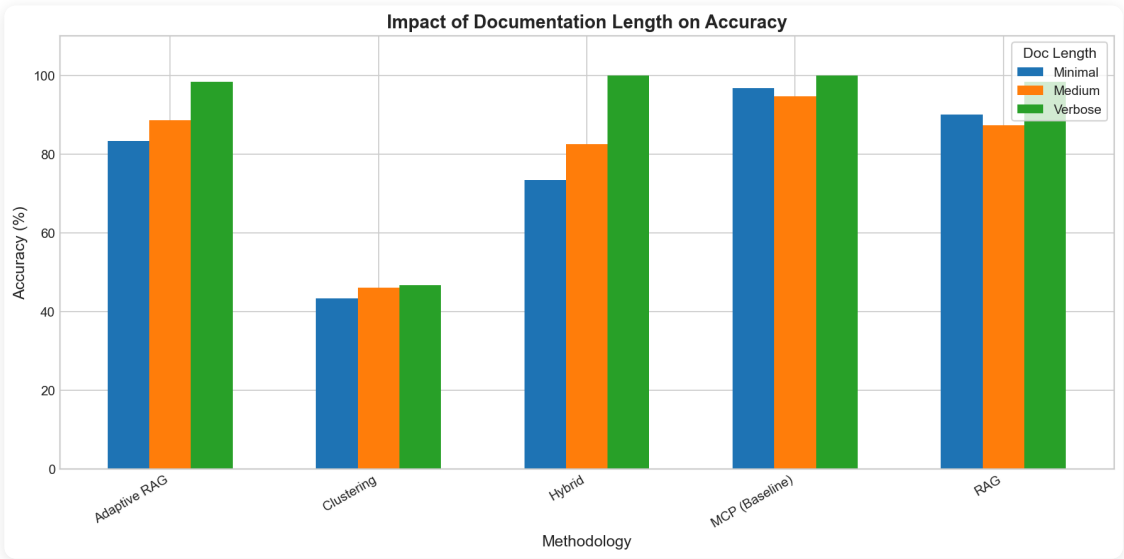
Latency distribution comparison. Left shows raw data with outliers, right shows filtered data.



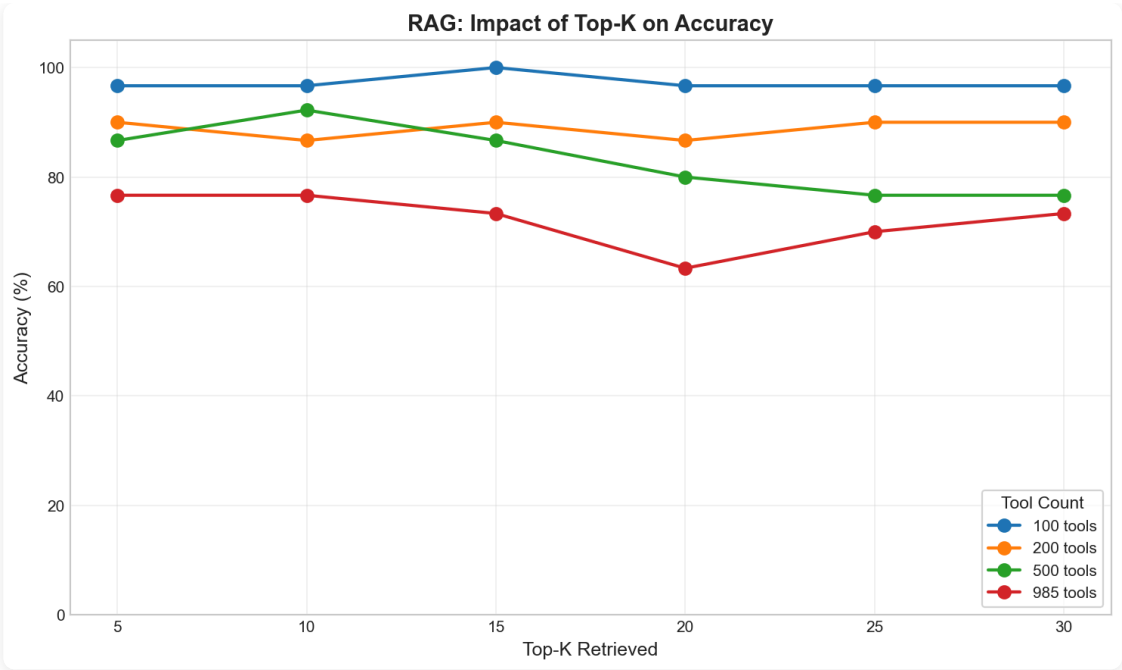
Trade-off between latency and accuracy. Point size indicates number of tools.



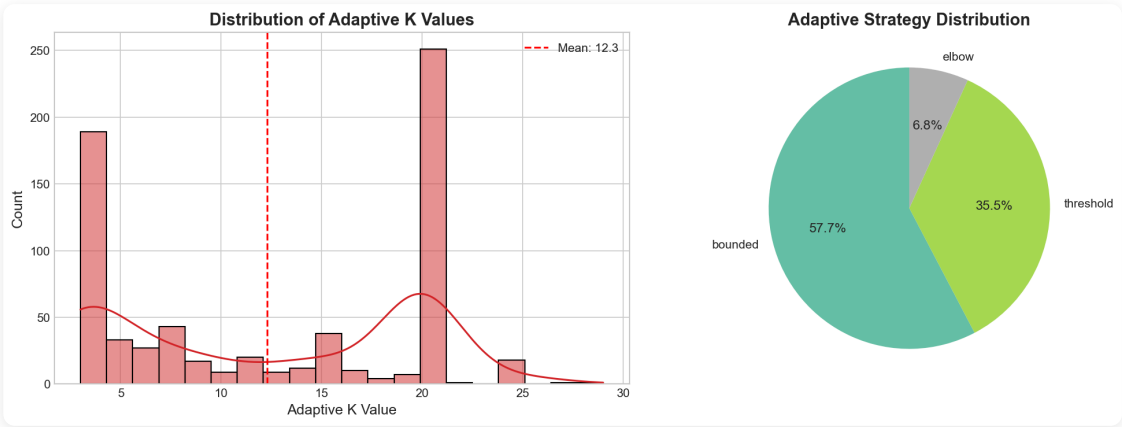
Breakdown of test outcomes by methodology.



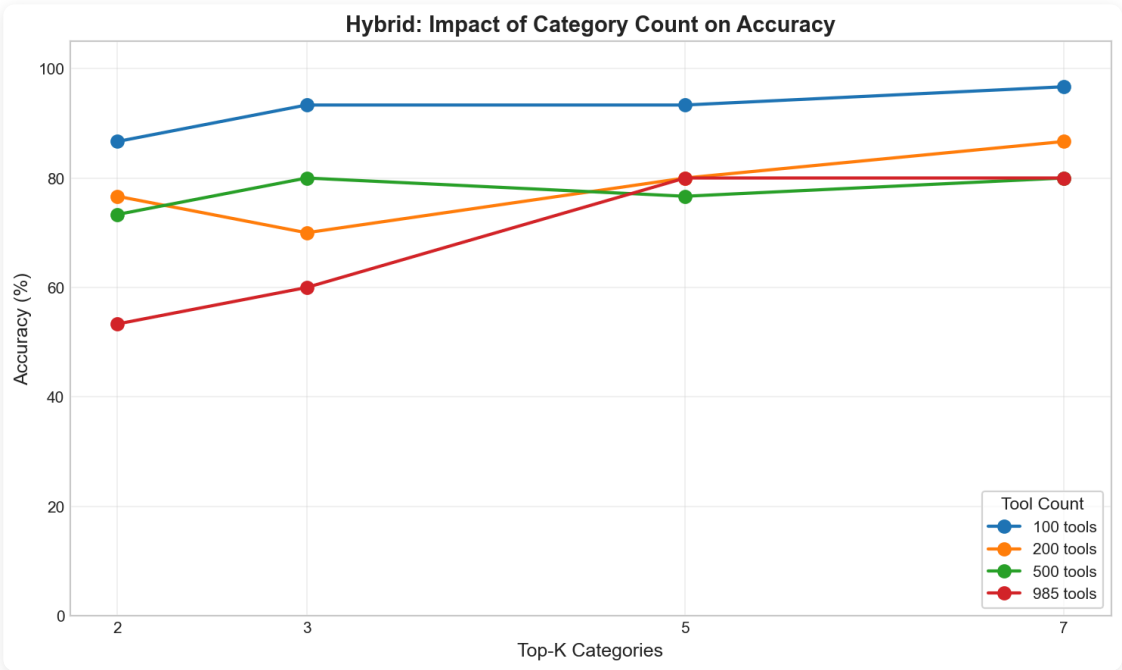
Impact of tool documentation length on accuracy.



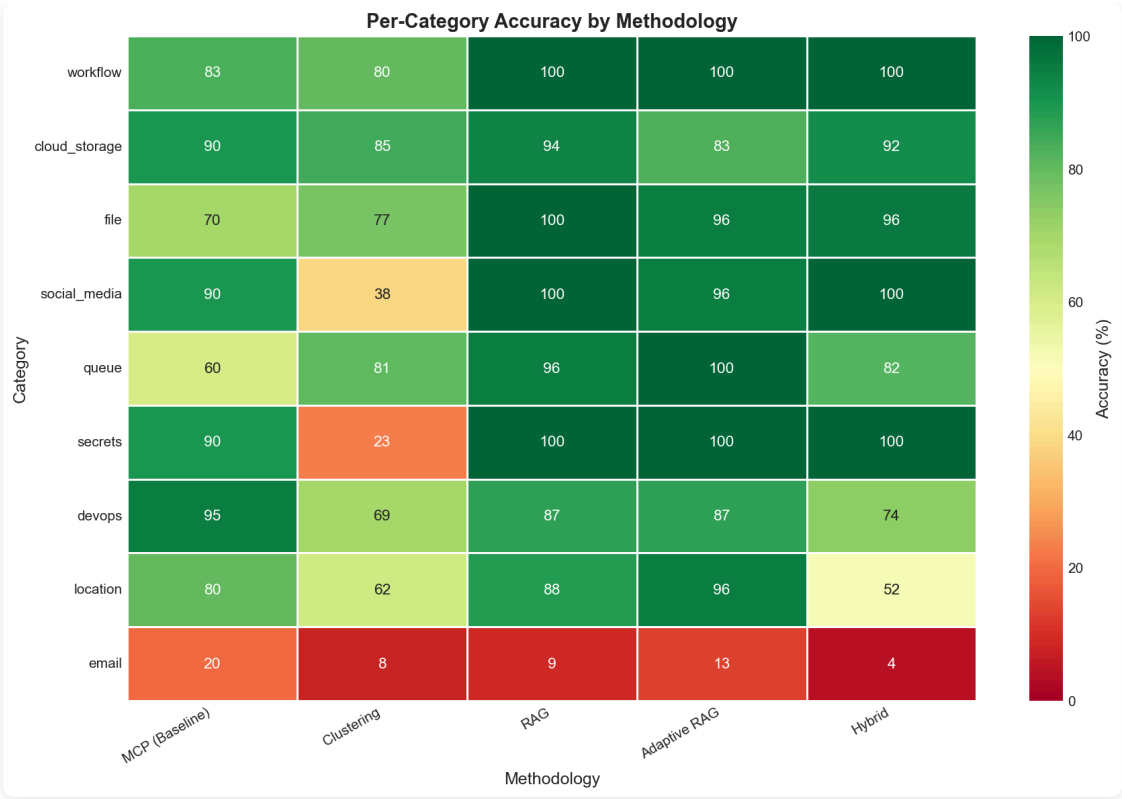
RAG methodology: Effect of top-K parameter on accuracy.



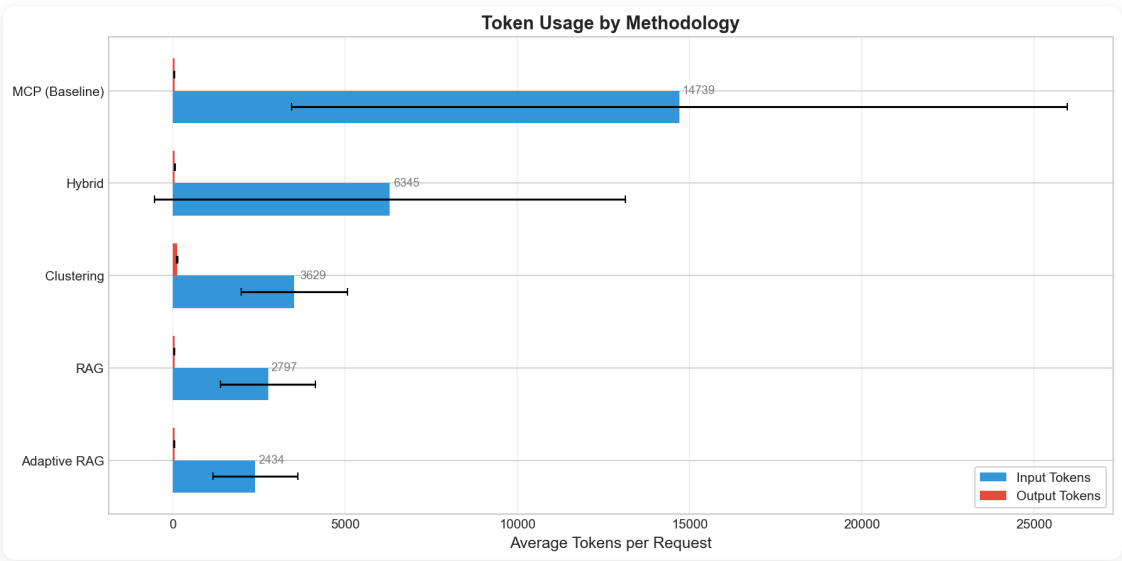
Adaptive RAG: Distribution of dynamically selected K values.



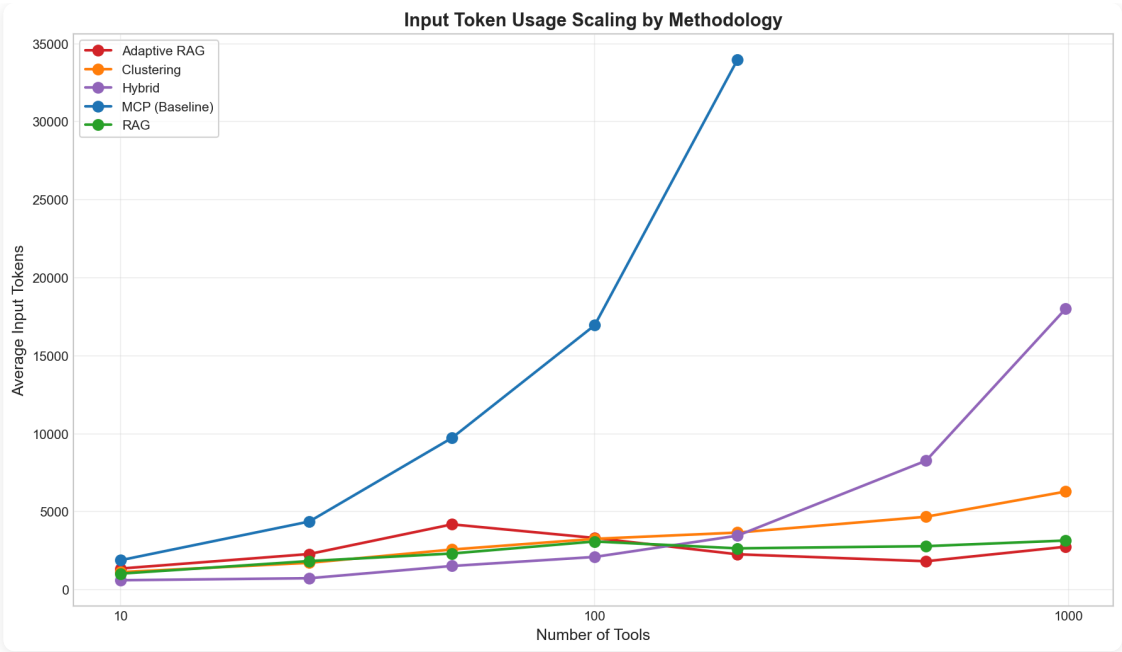
Hybrid methodology: Effect of top-K categories on accuracy.



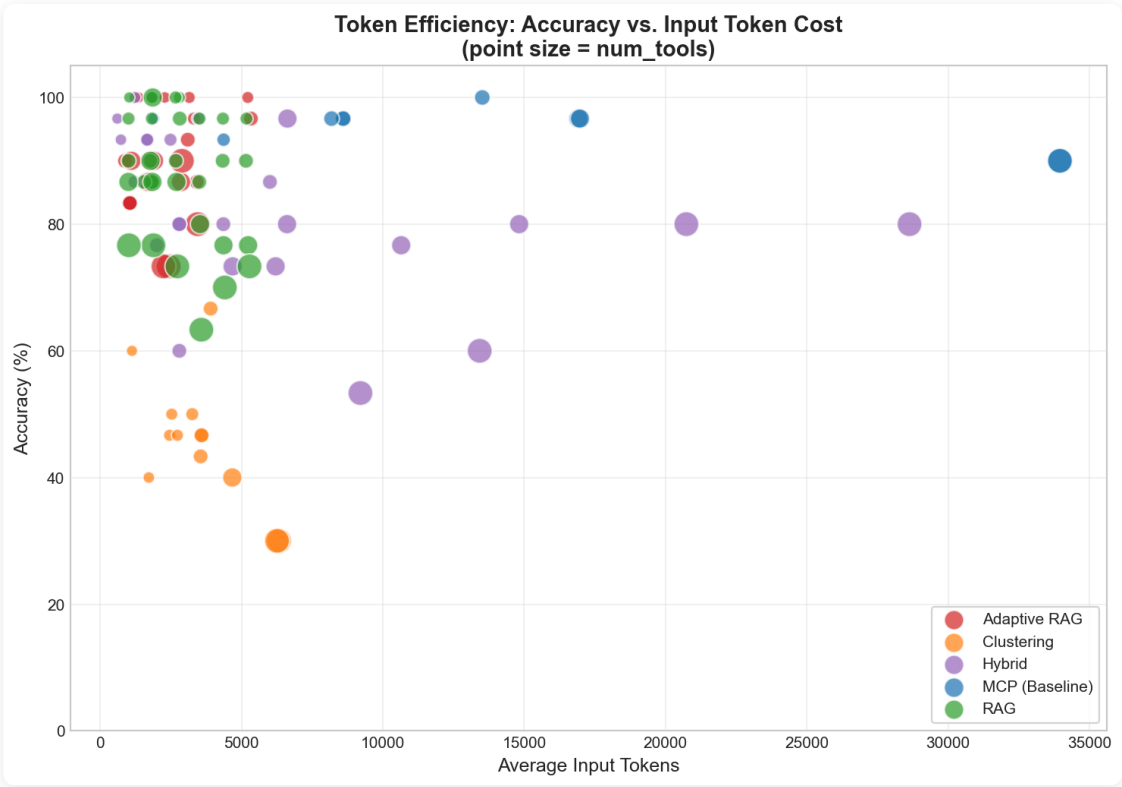
Per-category accuracy comparison across methodologies.



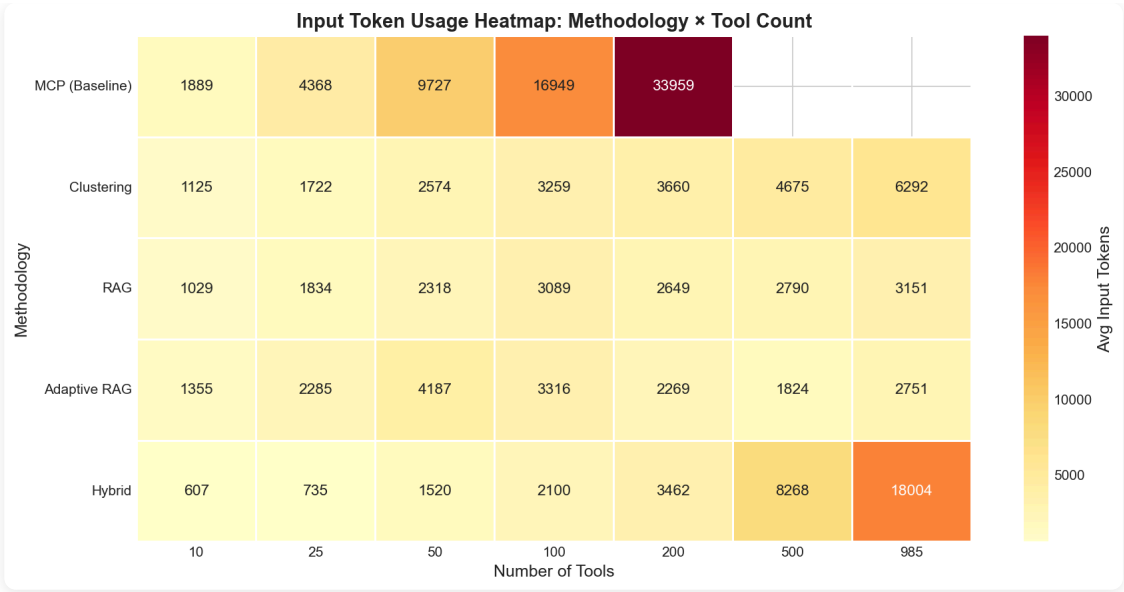
Average input and output tokens per request by methodology.



How input token usage scales with the number of tools for each methodology.



Token efficiency: accuracy achieved relative to input token cost.



Input token usage heatmap by methodology and tool count.

Key Findings

- **Best performing methodology:** Adaptive RAG with 93.1% average accuracy
- **MCP baseline:** Ranked #4 with 67.7% accuracy
- **Best at scale (500+ tools):** Adaptive RAG

Limitations & Notes

- Results may be affected by rate limiting during experiments
- Latency outliers have been filtered using IQR method in some charts
- Category accuracy may vary based on category representation in test set